

Daifeng Li

📍 Hefei, China ✉ fengli1702@gmail.com ☎ +86 152 4320 3967 🌐 github.com/fengli1702

EDUCATION

University of Science and Technology of China (USTC) **Hefei, China**
B.S. in Computer Science **Sep. 2022 – July 2026 (Expected)**

- **Honors Class:** Huaxia Computer Talent Program (Elite CS Program)
- **GPA:** 3.9/4.3 | **Weighted Avg:** 90.74/100 | **Rank:** 13/218 (Top 6%)
- **Selected Coursework:** Operating Systems (Honors, 99/100, **Top 1**), Computer Organization (Honors, 98/100), Algorithms (96/100), Stochastic Processes (96/100), Mathematical Logic (96/100).

RESEARCH INTERESTS

Machine Learning Systems (MLSys), GPU/CUDA Programming, Compiler and Programming Systems

PROJECT EXPERIENCE

Compiler-System Co-Design for GPU: IR-Level Optimizations **Summer 2025**
(Ongoing)

Research Assistant | Relaxed System Lab (HKUST)

- Developing compiler-system co-design strategies to optimize GPU kernel execution efficiency for Large Language Models.
- Implementing IR-level optimizations and abstracting CUDA memory hierarchy using **Tilelang** and **Triton** DSL.
- Leading code development, performance tuning, and evaluation of kernel fusion techniques on AMD and NVIDIA GPUs.

Revisiting Fairness in Cognitive Diagnosis: A Ranking-based Approach **Spring 2025**

Co-Author | Fairness Framework

- Proposed a novel ranking-based loss function to minimize the discrepancy between ideal and predicted rankings in cognitive diagnosis models.
- Implemented the training pipeline in PyTorch and conducted extensive experiments across 7 datasets.
- Achieved a 10% improvement in fairness metrics and significantly reduced overall loss; validated the approach on IRT models with optimal hyperparameters ($\lambda = 0.2$).

National Student Computer System Capability Challenge (NSCSCC) **Fall 2024**

Core Developer | National Third Prize Team

- Designed a high-performance **4-issue out-of-order Loongson CPU core** on FPGA.
- Implemented key micro-architectural components including the Reorder Buffer (ROB), Branch Prediction Unit, and non-blocking cache hierarchy.
- ✓ Developed CPU architecture, including out-of-order queues, computational units, and multiplication/division units

SKILLS

Languages: C, C++, Python, Verilog, Chisel, x86 Assembly, CUDA.

Frameworks & Tools: PyTorch, Triton, Git, GDB, $\LaTeX$, Linux.

TEACHING EXPERIENCE

- **Teaching Assistant**, Operating Systems (Honors), USTC Spring 2025
- **Teaching Assistant**, Computer Programming, USTC Fall 2025
- **Teaching Assistant**, Computer Programming, USTC Fall 2024

HONORS & AWARDS

- **National Scholarship** (Top 1%, Highest Honor from Ministry of Education), China 2025
- **National Scholarship** (Top 1%, Highest Honor from Ministry of Education), China 2024
- **Soong Ching Ling Future Scholarship** 2024
- **Soong Ching Ling Future Scholarship** 2023
- **National Third Prize**, National Student Computer System Capability Challenge 2024
- **Endeavor Scholarship**, USTC 2023